

In this analysis, we dive into **vulnerability-weighted event types**, a metric that combines media coverage frequency with **socioeconomic signals** like population density, poverty rates, and infrastructure resilience to prioritise climate risks. We leverage the massive **GDELT (Global Database of Events, Language, and Tone)** dataset: a real-time archive of 30+ years of global news articles processed via natural language processing to quantify mentions of extreme climate events (e.g., floods, heatwaves, droughts). This is cross-referenced with **government ground-truth datasets** from sources like India's National Disaster Management Authority, capturing verified metrics such as fatalities, economic damage, and affected populations.

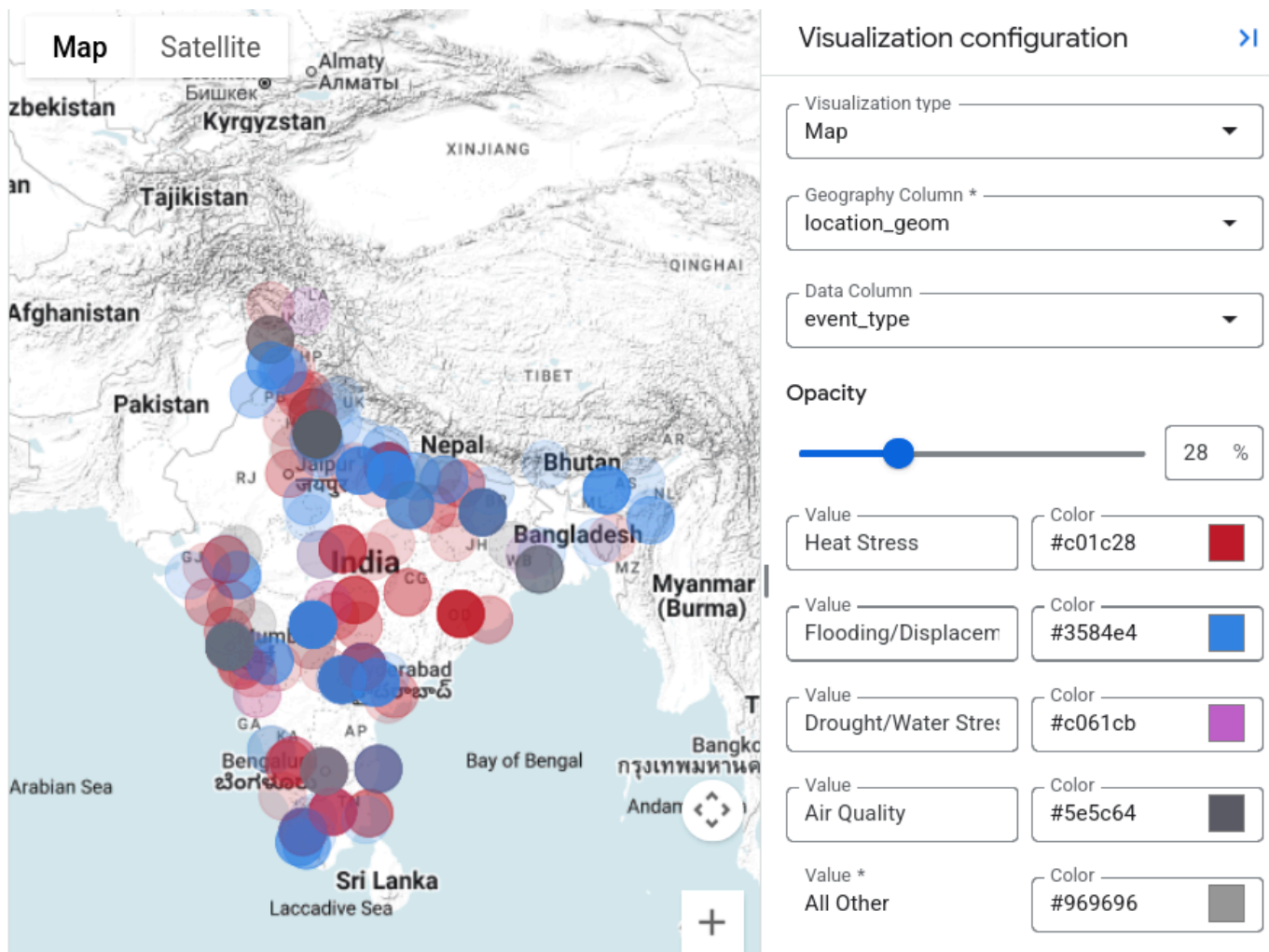
Queries run on **Google BigQuery** enable scalable queries across these **terabyte-scale datasets**, providing a data-driven visual lens to understand India's climate. This matters globally, as biased narratives shape policy funding which in-turn impacts us all.

Using GDELT data, we plot these events on a map to identify **hotspots** for each event type, determined by the intensity of media coverage. The visualisation highlights stark regional differences: some states receive far more attention than others, raising questions about whether this reflects true risk levels or reporting biases.

India's climate and disaster data landscape is plagued by **sporadic coverage and pervasive missing values**, creating blind spots in risk assessment and policy-making. This is compounded by fragmented datasets across states, where rural areas suffer from gaps in ground sensors, population metrics, and economic impact records. The **National Data & Analytics Platform (NDAP)** aims to address this by standardizing access to government data, but implementation lags reveal persistent issues.

Data Sources used:

- **Vulnerability Indicators:** NDAP dataset (district-level metrics like exposure, sensitivity, adaptive capacity), covering India's ~700 districts.
- **Media Intensity (GDELT):** Query GDELT's GEO 2.0 or DOC 2.0 API for April 2026 articles on, filtered by district names or ADM1/ADM2 codes. Aggregated by article counts and tone scores per district

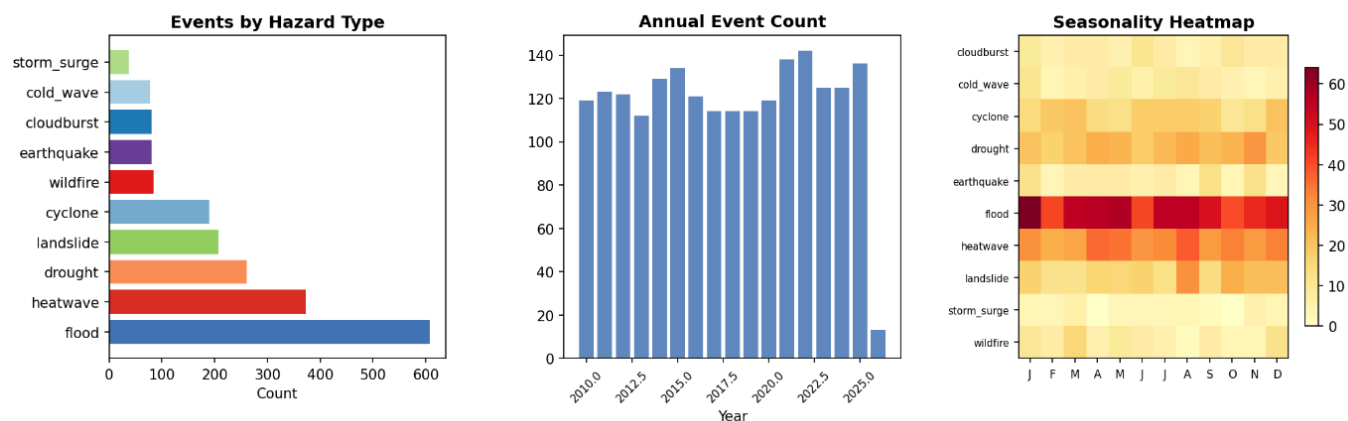


the GDELT dataset is used to infer how often extreme climate events such as floods, heatwaves, droughts, etc. appear in news coverage.

Government datasets provide the *ground truth*- fatalities, economic damage, or affected populations.

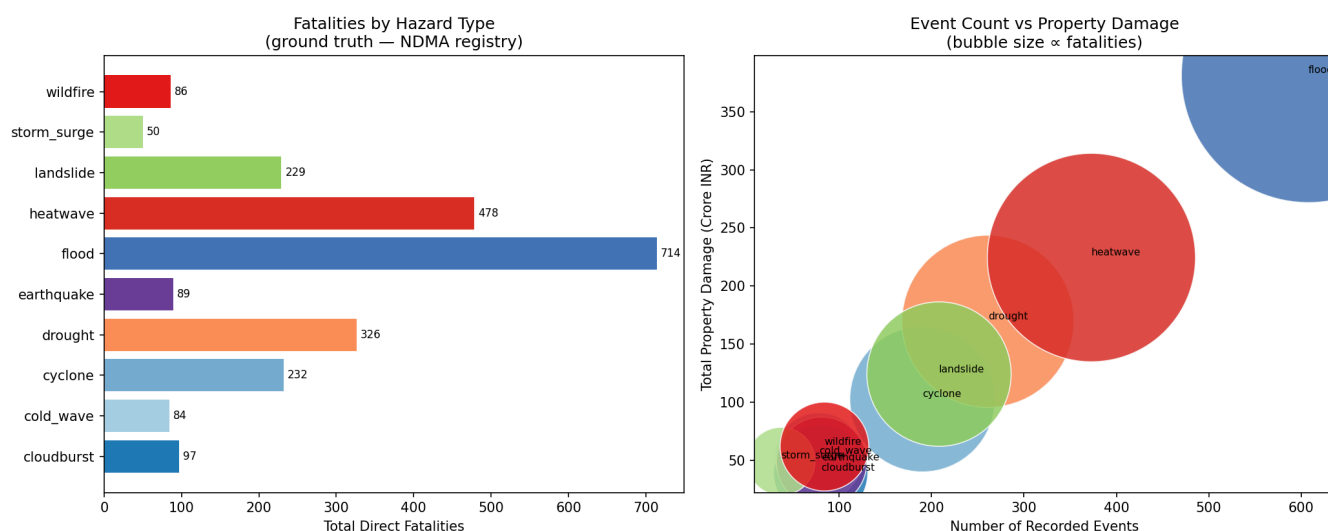
Putting them together reveals systematic distortions: fast, visible disasters like floods and cyclones tend to dominate coverage, while slow, cumulative risks like heat stress or drought often disappear from the narrative despite higher real-world impact until the stresses become unbearable.

India Climate Vulnerability — Dataset Overview NDMA Hazard Registry | 2010-2026



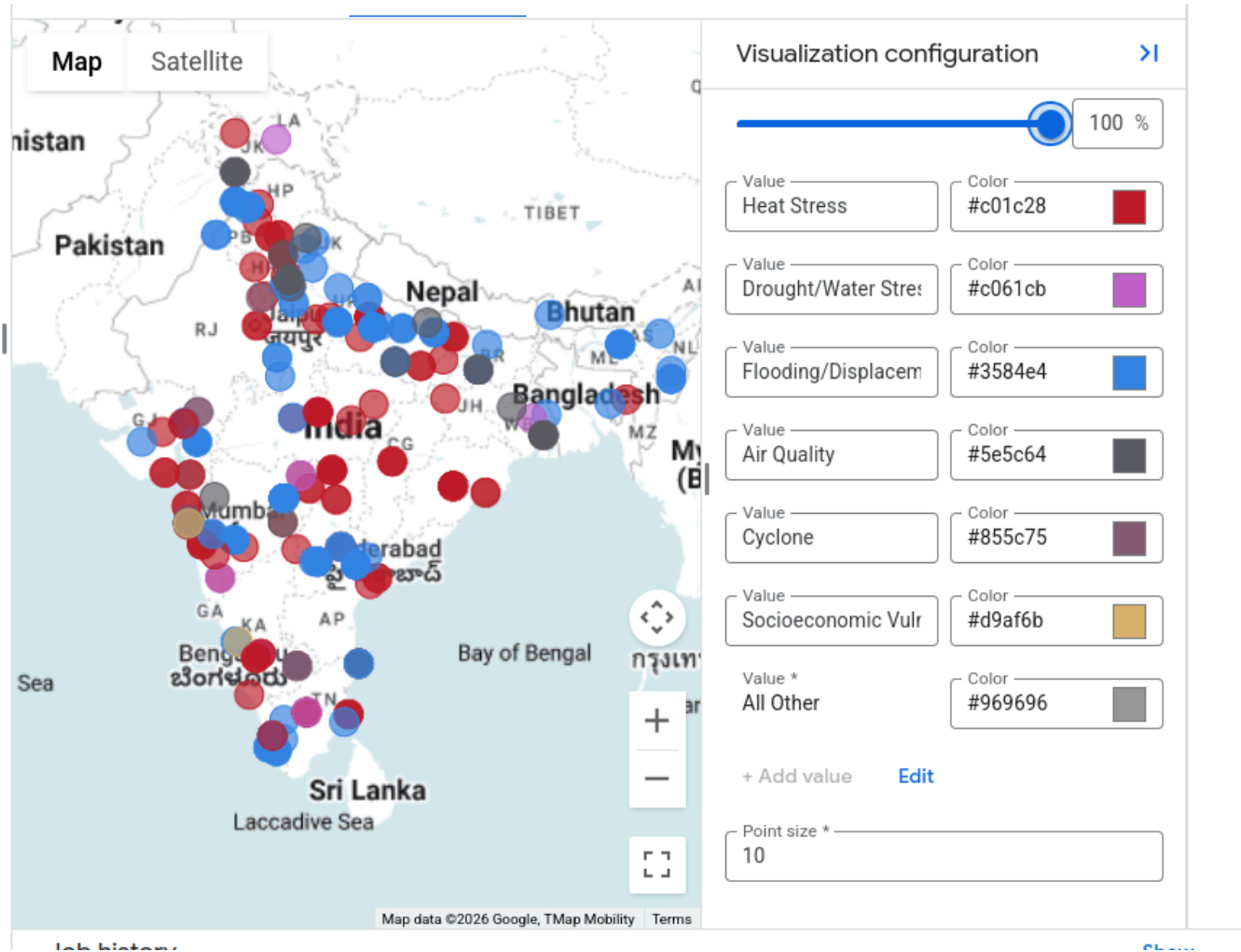
A quantitative analysis of the number and frequency of events shows us that India is extremely susceptible to floods and heatwaves.

RQ1 — Hazard Ground-Truth Impact Distribution

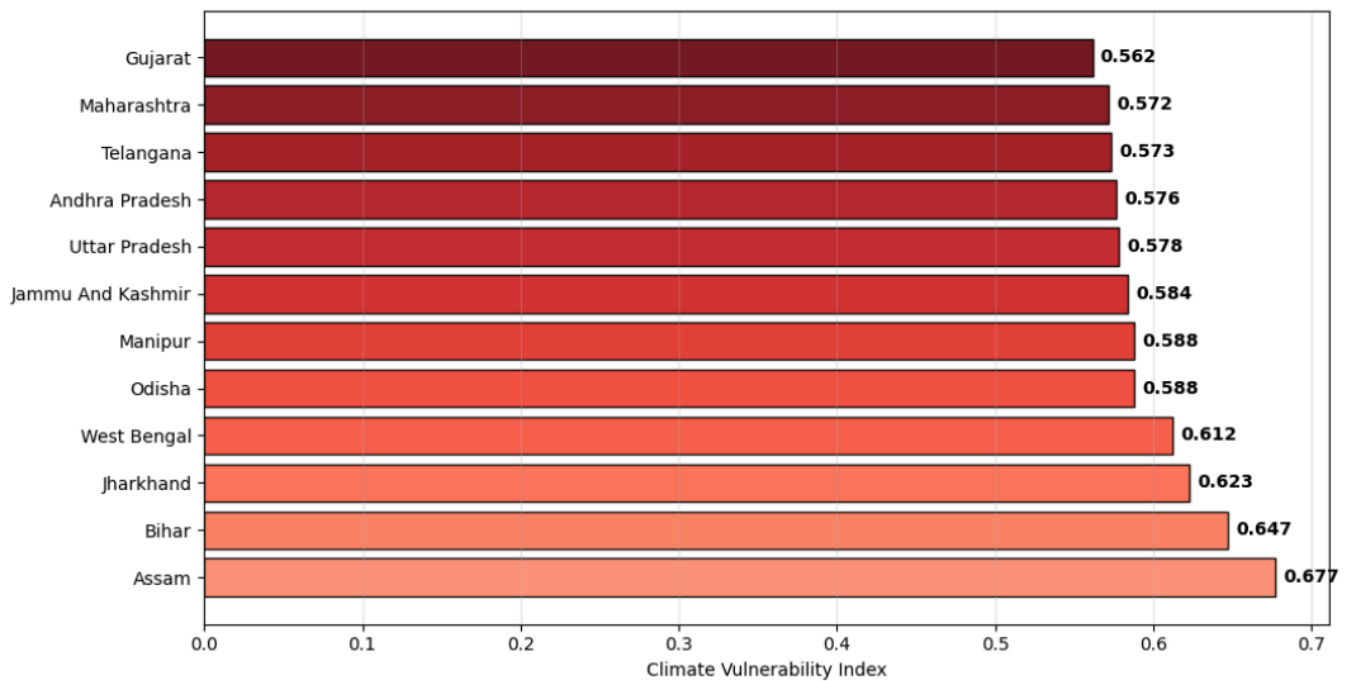


Using GDELT data to plot these events on a map show the hot-spots for each type of event based on analysis of media coverage. We notice some states have much higher coverage

than others.



To test if lower coverage signals under-reporting or genuinely lower vulnerability, we overlay socioeconomic factors. The results show clear discrepancies: issues in states like **Andhra Pradesh, Telangana, and Assam** appear under-reported relative to their risks. Additionally, socioeconomic dimensions receive insufficient attention in the media narrative.



We already see that issues from Andhra Pradesh, Telangana, and Assam are under-reported, and that there isn't enough reporting on socio-economic factors.

This analysis demonstrates how fusing GDELT media signals with NDAP socioeconomic indicators and NDMA ground-truth data via BigQuery expose distortions in India's climate coverage. Maps and charts reveal floods and heatwaves as media-dominant risks, while under-reported states like Andhra Pradesh, Telangana, and Assam show elevated vulnerabilities masked by sporadic data gaps.

Future work could refine NDAP integrations for real-time monitoring and expand to sub-district granularity, turning these insights into actionable tools for balanced policy prioritisation.